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CALPHAD: Computer Coupling of Phase Diagrams and Thermochemistry

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Thermodynamic modeling of the Co–Sm system

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ABSTRACT

The phase diagram of the Co–Sm system has been assessed by applying the *Calculation of Phase Diagram* (CALPHAD) technique. Thermochemical and phase equilibrium information from the literature have been critically evaluated, and a series of self-consistent thermodynamic parameters capable of describing the Gibbs free energies of each phase of the system has been obtained. There are eight compounds in this system. Based on an analysis of experimental data and the crystal structures of the phases, $SmCo_5$ was modeled as $(Co, Va)_{0.833333}(Co_2, Sm)_{0.166667}$ and Sm_2Co_{17} as $(Co)_{0.833333}(Sm)_{0.111111}(Co_2, Sm)_{0.055556}$. The remaining six intermetallic phases, Sm_3Co, Sm_9Co_4, $SmCo_2$, $SmCo_3$, Sm_2Co_7, and Sm_5Co_{19}, were treated as stoichiometric compounds. Calculations based on these parameters can reproduce most of the experimental data very well.

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1. Introduction

Alloys of the Co–Sm system have attracted much attention [1–13] for their excellent magnetic properties. Schneider et al. [14] and Su et al. [15] have thermodynamically assessed the Co–Sm system. However, Schneider did not give the final thermodynamic parameters in the publication [14], and for this reason, Su et al. [15] re-optimized the system. Recently, there have been studies that have rekindled some questions related to this system, such as unrealistic equilibrium and magnetic energy calculations [16,17] that have been often overlooked in the early literature. Such problems also exist in the work by Su et al. [15]. The use of thermodynamic parameters reported by Su et al. [15] for phase diagram calculations results in the appearance of an unrealistic inverted liquid miscibility gap at high temperatures. In addition, as pointed out by Ansara and Sundman [17], the calculation of the magnetic contribution to the Gibbs free energies should be modified. Following Su's work, there have been new experimental studies [18–21] in which Yuan et al. [19] have carried out a significant modification of the phase equilibria of the Sm-rich side. Stadelmaier et al. [22] have reviewed the Co–Sm system and given a detailed account of the structure and solubility ranges of the Co–Sm phases, especially for the $SmCo_5$ and Sm_2Co_{17} phase. Therefore, the aim of this work is to thermodynamically re-evaluate the Co–Sm system through application of the CALPHAD method using the available phase diagram and thermodynamic data, where the solubility of $SmCo_5$ and Sm_2Co_{17} are considered for the first time.

2. Literature information

2.1. Phase diagram data

The Co–Sm system was first fully investigated by Buschow et al. [1,4], who found seven intermetallic phases, namely Sm_3Co, Sm_9Co_4, $SmCo_2$, $SmCo_3$, Sm_2Co_7, $SmCo_5$ and Sm_2Co_{17} and constructed the phase diagram for the whole composition range. According to Buschow's results, Sm_3Co and Sm_2Co_{17} melt congruently and all other compounds are formed by peritectic reactions. $SmCo_5$ undergoes eutectoidal decomposition into Sm_2Co_7 and Sm_2Co_{17} at about 1023 K. Later studies revealed the existence of the Sm_5Co_{19} phase [11–13], having a similar structure to Sm_2Co_7 and formed by peritectic reaction. They attributed the absence of Sm_5Co_{19} reported in the earlier literature to its slow formation. It has been verified that the $SmCo_5$ phase is unstable [8–12] at low temperature and decomposes into Sm_5Co_{19} and Sm_2Co_{17} *via* a eutectoid reaction [10–12]. Ge et al. [12] re-investigated the Co–Sm system fully and achieved a result similar to that of Buschow et al. [1,4] excluding the part related to Sm_5Co_{19}. Ge et al. [12] reported the decomposition of $SmCo_5$ taking place at 1083 K and found a small solubility of Sm in fcc-Co. The temperature of the invariant reactions on the Co-rich side reported

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in [1,12] are very close, but those reported by Cataldo et al. [13] are generally lower. Yuan et al. [19] re-investigated the Co–Sm system on the Sm-rich side. The phase relationships between the three allotropes of Sm and Sm_3Co have been clarified, and the Sm_9Co_4 phase has been revised as having the Sm_5Co_2 stoichiometry. Therefore, in this work, the phase diagram data reported by [1,4,12,19] have been given higher weights in the optimization.

As $SmCo_5$ and Sm_2Co_{17} are the major phases in Co–Sm permanent magnets, this has led to the Co-rich side of Co–Sm system being investigated extensively [5,6,12,13,22]. It is generally accepted that the homogeneity range of $SmCo_5$ extends to both the Co-rich and Sm-rich side of the stoichiometric composition and that of Sm_2Co_{17} extends only to Sm-rich side. The crystal structure of the $SmCo_5$ phase is of the $CuZn_5$ type and the Sm_2Co_{17} phase is of Th_2Ni_{17} type at high temperatures. Sm_2Co_{17} can be transformed from the $SmCo_5$ structure, where each third Sm atom in the basal plane is substituted by a pair of dumbbell Co atoms [2]. Stadelmaier et al. [22] pointed out that the Sm-rich $SmCo_5$ is probably derived from the replacement of Co sites by a vacancy; the Co-rich $SmCo_5$ from a random substitution of Sm atoms with a Co pair (dumbbells); and the Sm-rich Sm_2Co_{17} from the replacement of Co pairs by Sm again.

2.2. Thermochemical data

Berezutskii and Ivanov [21] have measured the partial mixing enthalpies of Co–Sm liquid alloys by high-temperature calorimetry and their results have been used in this optimization.

The limiting partial enthalpy of mixing in Co–Sm liquid alloys as measured by Usenko et al. [20] are used to check the optimized result.

The value of enthalpies of formation of the Co–Sm compounds reported by [23,24] have shown large discrepancies and both were given lower weights in the optimization.

2.3. Magnetic data

Buschow [25] compiled the magnetic data of various compounds related to rare-earth metals. Campos et al. [26] re-measured the values of Curie temperature (T_C) of Sm_2Co_7, Sm_5Co_{19}, $SmCo_5$ and Sm_2Co_{17}. In the calculation of the magnetic contribution to Gibbs free energies, the magnetic moment was incorporated into the Gibbs energy expressions by the number of sites of the magnetic constituents, as illustrated by Ansara and Sundman [17]. Since Sm and Co atoms in Co–Sm magnetic phases are coupled and both elements contribute to the magnetic moment [27], the mean value of the magnetic moment in per mole atom has been adopted for simplicity, the difference between their individual contributions not being considered.

3. Thermodynamic models

3.1. Liquid, fcc, hcp, bcc, and rhombohedral phases

In this work, the Gibbs free energies of pure Co and Sm were taken from the SGTE pure-element database [28]. Pure Sm is treated as non-magnetic because it only shows magnetic features at very low temperatures.

Liquid, fcc, hcp, bcc and rhomb phases are modeled as substitutional solutions. The model assumes random mixing in one sublattice and is described as:

$$G_m^{\phi} = x_{Co}\,{}^0G_{Co}^{\phi} + x_{Sm}\,{}^0G_{Sm}^{\phi} + RT(x_{Co}\ln x_{Co} + x_{Sm}\ln x_{Sm}) + {}^{ex}G_m^{\phi} + {}^{mg}G_m^{\phi} \quad (1)$$

where x_i is the mole fraction of element i, ${}^0G_{Sm}^{\phi}$ and ${}^0G_{Co}^{\phi}$ are the molar Gibbs energies of the respective pure elements with structure ϕ, R is gas constant, T is the absolute temperature, and ${}^{ex}G_m^{\phi}$ is the excess Gibbs energy of phase ϕ expressed in the form of Redlich–Kister polynomials:

$${}^{ex}G_m^{\phi} = x_{Co}x_{Sm}\sum_{i=0}^{n}{}^iL_{Co,Sm}^{\phi}(x_{Co} - x_{Sm})^i \quad (2)$$

in which ${}^iL_{Co,Sm}^{\phi}$ is the binary interaction parameter of the corresponding phase to be optimized from experimental data. ${}^{mg}G_m^{\phi}$ is the magnetic contribution to the Gibbs energy and will be discussed later.

The lattice stability of Co in the rhombohedral structure is given as ${}^0G_{Co}^{rho} = {}^0G_{Co}^{hcp} + 2500\ \mathrm{J\ mol^{-1}}$.

3.2. Stoichiometric intermetallic compounds

The intermetallic compounds, Sm_3Co, Sm_9Co_4, $SmCo_2$, $SmCo_3$, Sm_2Co_7 and Sm_5Co_{19}, in the Co–Sm system are treated as stoichiometric phases. Their Gibbs energies per mole of atoms are expressed as:

$$G_m^{Sm_aCo_b} = a\,{}^0G_{Sm}^{rho} + b\,{}^0G_{Co}^{rho} + A^{Sm_aCo_b} + B^{Sm_aCo_b}T + {}^{mg}G_m^{Sm_aCo_b} \quad (3)$$

where ${}^0G_{Sm}^{rho}$ and ${}^0G_{Co}^{rho}$ are the molar Gibbs energies of the respective pure elements. $A^{Sm_aCo_b}$ and $B^{Sm_aCo_b}$ are the parameters to describe the Gibbs energy of formation and are optimized using experimental data. ${}^{mg}G_m^{Sm_aCo_b}$ is the magnetic contribution to Gibbs energy and will be discussed in Section 3.4.

3.3. $SmCo_5$ and Sm_2Co_{17} phases

The structural transformation of the Sm_2Co_{17} phase is not considered in this assessment. It is commonly accepted that both the $SmCo_5$ and Sm_2Co_{17} phases have some solubility at high temperatures. In this work, the models of the $SmCo_5$ and Sm_2Co_{17} phases are constructed on the basis of the substitution mechanisms outlined by Stadelmaier et al. [22]. Co pairs have been incorporated into the model as a specie as Co_2 [29]. The $SmCo_5$ phase is formulated as $(Co, Va)_{0.833333}(Co_2, Sm)_{0.166667}$ with solubility on both sides of the ideal stoichiometry, and the Sm_2Co_{17} phase is treated as $(Co)_{0.833333}(Sm)_{0.111111}(Co_2, Sm)_{0.055556}$. The full expression of Gibbs free energy for these phases is described as:

$$\begin{aligned} G_m^{SmCo_5} = {} & y'_{Co}y''_{Co_2}G_{Co:Co_2} + y'_{Co}y''_{Sm}G_{Co:Sm} \\ & + y'_{Va}y''_{Co_2}G_{Va:Co_2} + y'_{Va}y''_{Sm}G_{Va:Sm} \\ & + 0.833333 \times RT\left(y'_{Co}\ln y'_{Co} + y'_{Va}\ln y'_{Va}\right) \\ & + 0.166667 \times RT\left(y''_{Co_2}\ln y''_{Co_2} + y''_{Sm}\ln y''_{Sm}\right) \\ & + y'_{Co}y'_{Va}y''_{Co_2}L_{Co,Va:Co_2} + y'_{Co}y'_{Va}y''_{Sm}L_{Co,Va:Sm} \\ & + y'_{Co}y''_{Sm}y''_{Co_2}L_{Co:Co_2,Sm} + y'_{Va}y''_{Sm}y''_{Co_2}L_{Va:Co_2,Sm} \end{aligned} \quad (4)$$

and

$$\begin{aligned} G_m^{Sm_2Co_{17}} = {} & y_{Co_2}\,{}^0G_{Co:Sm:Co_2} + y_{Sm}\,{}^0G_{Co:Sm:Sm} \\ & + 0.055556 \times RT\left(y_{Co_2}\ln y_{Co_2} + y_{Sm}\ln y_{Sm}\right) \\ & + y_{Sm}y_{Co_2}L_{Co:Sm:Co_2,Sm} \end{aligned} \quad (5)$$

respectively. Where y_i^n is the species concentrations of element i on the different sublattices with $y_{Sm}^n + y_{Co}^n = 1$.

The Gibbs energy value of each end member is described as:

$$G_{Sm_xCo_y} = x\,{}^0G_{Sm}^{rho} + y\,{}^0G_{Co}^{hcp} + A^{Sm_xCo_y} + B^{Sm_xCo_y}T + {}^{mg}G_m^{Sm_xCo_y} \quad (6)$$

where ${}^{mg}G_m^{Sm_xCo_y}$ is the magnetic contribution to Gibbs energy.

Table 1
The values of T_C, β and n adopted in the optimization of Sm–Co system. The values are given in SI units per mole of atoms.

Phase	$SmCo_2$	$SmCo_3$	Sm_2Co_7	Sm_5Co_{19}	$SmCo_5$	Sm_2Co_{17}
T_C (K)	226.7	533	705	786	957	1173
β	0.44	0.775	1.01	1.10	1.22	1.43
n	1	1	1	1	1	1

Phase	$(Co)_{0.833333}:(Co_2)_{0.166667}$	$(Va)_{0.833333}:(Co_2)_{0.166667}$
T_C (K)	1396	1396
β	1.71	0.33
n	1.166667	0.333334

3.4. Magnetic effects

As expressed by Hillert and Jarl [30], the magnetic contribution to the Gibbs energy is given as:

$$^{mg}G_m^{\phi} = RT\ln(\beta+1)f(\tau) \tag{7}$$

where τ is defined as T/T_C, β is a constant related to the total magnetic entropy and $f(\tau)$ represents a polynomial assumed by Hillert and Jarl [30] as:

$$f(\tau) = 1 - \left[\frac{79\tau^{-1}}{140p} + \frac{474}{497}\left(\frac{1}{p}-1\right)\left(\frac{\tau^3}{6}+\frac{\tau^9}{135}+\frac{\tau^{15}}{600}\right)\right]\Big/D \quad \text{for } \tau \le 1 \tag{8}$$

$$f(\tau) = -\left(\frac{\tau^{-5}}{10}+\frac{\tau^{-15}}{315}+\frac{\tau^{-25}}{1500}\right)\Big/D \quad \text{for } \tau > 1 \tag{9}$$

where $D = \frac{518}{1125} + \frac{11692}{15975}\left(\frac{1}{p}-1\right)$, and p depends on the structure, $p = 0.4$ for bcc structure and 0.28 for others.

β has the same value as the Bohr magnetic moment per mole of atoms. T_C values for $SmCo_2$ and $SmCo_3$ are taken from [25]. The most recent values of T_C for Sm_2Co_7, Sm_5Co_{19}, $SmCo_5$ and Sm_2Co_{17} reported by Campos et al. [26] are adopted. Ansara and Sundman [17] proposed that the magnetic moments, β, should be expressed in per one sublattice site for the magnetic element, and corresponding the magnetic contributions to Gibbs energy should be calculated in the same manner. Since the 'β' value in [25] is in Bohr magnetic moment per formula unit, β = 'β'/ $(a_{Sm} + b_{Co})$ was used here and energy calculations are formulated per mole of atoms. It should be noted that the difference in the magnetic contributions of Sm and Co atoms has been ignored in the calculation. Both Sm_3Co and Sm_4Co_9 are treated as non-magnetic phases in this work. The T_C and β values are listed in Table 1.

In the calculation of the magnetic contribution to the Gibbs energies of the end members, $(Co)_{0.833333}:(Co_2)_{0.166667}$ and $(Va)_{0.833333}:(Co_2)_{0.166667}$, for the $SmCo_5$ phase, the 'β' value of hcp-Co was adopted and transformed into $\beta = (\text{'}\beta\text{'}+1)^n - 1$, where $n = 1.166667$ and 0.333334 respectively, as shown in Table 1. The related T_C values are taken from hcp-Co, as listed in Table 1.

4. The optimization procedure

The thermodynamic parameters of the Co–Sm system were optimized using Pandat software [31]. By using a least squares method, the thermodynamic parameters were adjusted in order to minimize the sum of the squares of the errors between calculated and experimental phase equilibrium and thermodynamic data.

Firstly, the parameters of the liquid and terminal solid-solutions were optimized by using the liquid mixing enthalpies, and the liquidus and solidus phase boundary information. Secondly, all intermetallic phases, including $SmCo_5$ and Sm_2Co_{17} were treated as stoichiometric compounds. The intermetallic compounds were incorporated one by one from the Sm-rich side. In each step, the liquid parameters were modified according to invariant reaction points and the liquidus. After the entire phase diagram could be faithfully plotted and the phase transformation relations well represented, the solubility of the $SmCo_5$ and Sm_2Co_{17} phases were introduced and their parameters optimized with all the other parameters fixed. Finally, all the parameters for the complete system were evaluated together in order to give the optimal description of the full system.

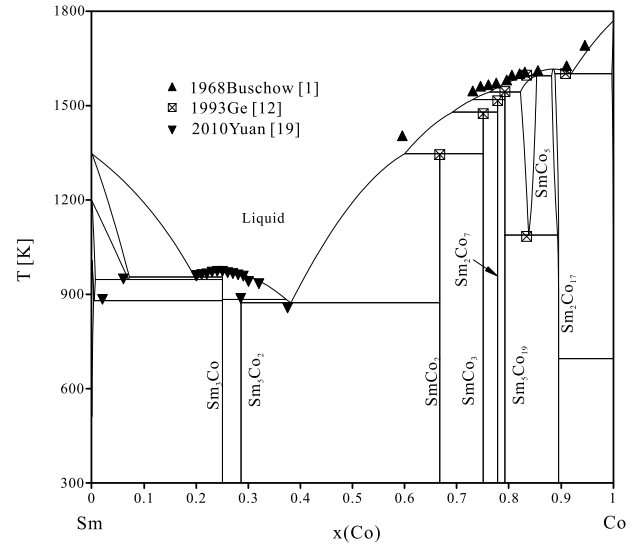

Fig. 1. The assessed Sm–Co system in this work.

Table 2
The invariant reactions in Sm–Co system.

Invariant reaction	T (K)			
	Present work	[19]	[12]	[1]
Liq ↔ γ(Sm) + Sm_3Co	954	958	–	–
γ(Sm) ↔ β(Sm) + Sm_3Co	946	948	–	–
β(Sm) ↔ α(Sm) + Sm_3Co	879	883	–	–
Liq ↔ Sm_3Co	970	968	973	968
Liq + Sm_3Co ↔ Sm_5Co_2	883	886	886	878
Liq ↔ Sm_5Co_2 + $SmCo_2$	872	856	863	848
Liq + $SmCo_3$ ↔ $SmCo_2$	1345	–	1343	1348
Liq + Sm_2Co_7 ↔ $SmCo_3$	1477	–	1473	1473
Liq + Sm_5Co_{19} ↔ Sm_2Co_7	1517	–	1514	1513
Liq + $SmCo_5$ ↔ Sm_5Co_{19}	1541	–	1543	–
Liq + Sm_2Co_{17} ↔ $SmCo_5$	1593	–	1595	1593
Liq ↔ Sm_2Co_{17}	1614	–	1620	1608
Liq ↔ α(Co) + Sm_2Co_{17}	1600	–	1600	1598
$SmCo_5$ ↔ Sm_5Co_{19} + Sm_2Co_{17}	1087	–	1083	–

5. Results and discussion

The calculated phase diagram of Co–Sm system is shown in Fig. 1. Fig. 2 is an enlarged view of the Co-rich side. Comparisons of the calculated invariant reactions with experimental data are listed in Table 2. Table 3 lists the optimized parameters.

As shown in Fig. 1 and Table 2, the calculated results agree well with experimental data. The homogeneity ranges of the $SmCo_5$ and Sm_2Co_{17} phases, together with literature data are presented in Fig. 2. The homogeneity ranges of $SmCo_5$ and Sm_2Co_{17} agree well with literature data.

Fig. 3 shows the calculated mixing enthalpy of liquid alloys and the partial mixing enthalpy for the liquid phase at 1829 K, compared with literature data [21]. The calculated limiting partial enthalpy of mixing of Co in the Co–Sm liquid at 1400 K is −55.5 kJ/mol, which agrees well with the value of −50.5 ± 3.9 kJ/mol reported by Usenko and Ivanov [20].

Table 3
The model parameters of Sm–Co system obtained in this work, given in SI units per mole of atoms.

Phases	Descriptions
Liquid	Sublattice model: (Co, Sm)
	${}^0L_{\mathrm{Co,Sm}} = -65705 + 12.386T$
	${}^1L_{\mathrm{Co,Sm}} = -10246 + 0.7777T$
Fcc_A1	Sublattice model: (Co, Sm) (Va)
	${}^0L_{\mathrm{Co,Sm:}Va} = -9958.34$
Hcp_A3	Sublattice model: (Co, Sm)$(\mathrm{Va})_{0.5}$
	${}^0L_{\mathrm{Co,Sm:}Va} = 0$
Bcc_A2	Sublattice model: (Co, Sm)$(\mathrm{Va})_3$
	${}^0L_{\mathrm{Co,Sm:}Va} = -24500$
Rhomb-C19	Sublattice model: (Co, Sm)
	${}^0L_{\mathrm{Co,Sm:}Va} = 0$
Sm_3Co	Sublattice model: $(\mathrm{Co})_{0.25}(\mathrm{Sm})_{0.75}$
	$G^{\mathrm{Sm_3Co}}_{\mathrm{Co:Sm}} = 0.75\,{}^0G^{rho}_{\mathrm{Sm}} + 0.25\,{}^0G^{hcp}_{\mathrm{Co}} - 10053.5 + 0.0407T$
Sm_5Co_2	Sublattice model: $(\mathrm{Co})_{0.285714}(\mathrm{Sm})_{0.714286}$
	$G^{\mathrm{Sm_9Co_4}}_{\mathrm{Co:Sm}} = 0.714286\,{}^0G^{rho}_{\mathrm{Sm}} + 0.285714\,{}^0G^{hcp}_{\mathrm{Co}} - 10748 + 0.1T$
$SmCo_2$	Sublattice model: $(\mathrm{Co})_{0.666667}(\mathrm{Sm})_{0.333333}$
	$G^{\mathrm{SmCo_2}}_{\mathrm{Co:Sm}}$
	$= 0.333333\,{}^0G^{rho}_{\mathrm{Sm}} + 0.666667\,{}^0G^{hcp}_{\mathrm{Co}} - 16827.2 - 0.2441T$
$SmCo_3$	Sublattice model: $(\mathrm{Co})_{0.75}(\mathrm{Sm})_{0.25}$
	$G^{\mathrm{SmCo_3}}_{\mathrm{Co:Sm}} = 0.25\,{}^0G^{rho}_{\mathrm{Sm}} + 0.75\,{}^0G^{hcp}_{\mathrm{Co}} - 16666.5 + 0.4851T$
Sm_2Co_7	Sublattice model: $(\mathrm{Co})_{0.777778}(\mathrm{Sm})_{0.222222}$
	$G^{\mathrm{Sm_2Co_7}}_{\mathrm{Co:Sm}}$
	$= 0.222222\,{}^0G^{rho}_{\mathrm{Sm}} + 0.777778\,{}^0G^{hcp}_{\mathrm{Co}} - 156124 + 0.32T$
Sm_5Co_{19}	Sublattice model: $(\mathrm{Co})_{0.791667}(\mathrm{Sm})_{0.208333}$
	$G^{\mathrm{Sm_5Co_{19}}}_{\mathrm{Co:Sm}}$
	$= 0.208333\,{}^0G^{rho}_{\mathrm{Sm}} + 0.791667\,{}^0G^{hcp}_{\mathrm{Co}} - 14865 + 0.16T$
$SmCo_5$	Sublattice model: $(\mathrm{Co}, Va)_{0.833333}(\mathrm{Co_2}, \mathrm{Sm})_{0.166667}$
	$G^{\mathrm{SmCo_5}}_{\mathrm{Co:Sm}} = 0.166667\,{}^0G^{rho}_{\mathrm{Sm}} + 0.833333\,{}^0G^{hcp}_{\mathrm{Co}} - 12680 + 1.72T$
	$G^{\mathrm{SmCo_5}}_{\mathrm{Co:Co_2}} = 1.166667\,{}^0G^{hcp}_{\mathrm{Co}} + 8500$
	$G^{\mathrm{SmCo_5}}_{Va\mathrm{:Sm}} = 1.166667\,{}^0G^{rho}_{\mathrm{Sm}} + 10910$
	$G^{\mathrm{SmCo_5}}_{Va\mathrm{:Co_2}} = 0.333334\,{}^0G^{hcp}_{\mathrm{Co}} + 10000$
	$L_{\mathrm{Co},Va\mathrm{:Co_2}} = -19980 + 11.3T$
	$L_{\mathrm{Co},Va\mathrm{:Sm}} = -19980 + 11.3T$
	$L_{\mathrm{Co:Co_2,Sm}} = -13750 + 5.9T$
	$L_{Va\mathrm{:Co_2,Sm}} = -13750 + 5.9T$
Sm_2Co_{17}	Sublattice model:
	$(\mathrm{Co})_{0.833333}(\mathrm{Sm})_{0.111111}(\mathrm{Co_2}, \mathrm{Sm})_{0.055556}$
	$G^{\mathrm{Sm_2Co_{17}}}_{\mathrm{Co:Sm:Co_2}}$
	$= 0.944445\,{}^0G^{rho}_{\mathrm{Sm}} + 0.111111\,{}^0G^{hcp}_{\mathrm{Co}} - 10070 + 0.4T$
	$G^{\mathrm{Sm_2Co_{17}}}_{\mathrm{Co:Sm:Sm}}$
	$= 0.166667\,{}^0G^{rho}_{\mathrm{Sm}} + 0.833333\,{}^0G^{hcp}_{\mathrm{Co}} - 12680 + 1.72T$
	$L_{\mathrm{Co:Sm:Co_2,Sm}} = 5536.2 - 5.5T$

Calculation of the Sm–Co phase diagram has been carried out up to a temperature of 6000 K and no miscibility gap or unrealistic phase equilibria were seen.

6. Summary

A self-consistent thermodynamic description of the Co–Sm system has been obtained by assessment of experimental data. The results agree well with literature information and reproduce all the phase relations in this system very well.

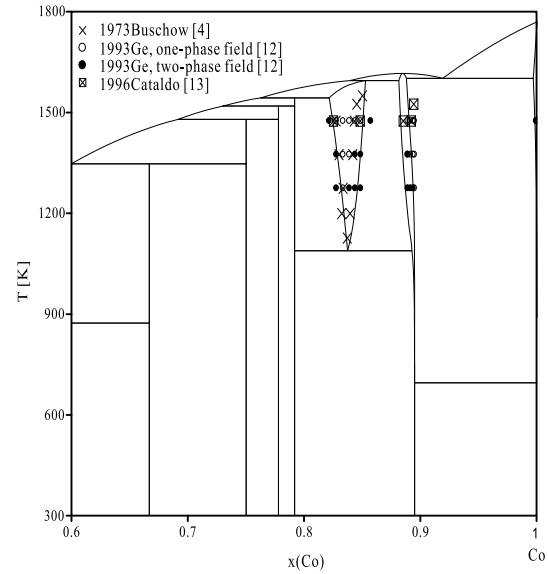

Fig. 2. The magnified region of Sm–Co phase diagram illustrating the liquidus and the homogeneity ranges of $SmCo_5$ and Sm_2Co_{17}.

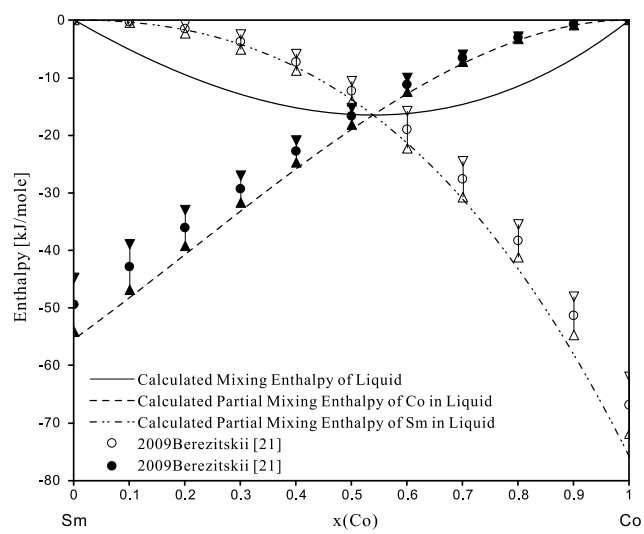

Fig. 3. Calculated enthalpies of mixing of Sm–Co liquid and related partial mixing enthalpies at 1829 K and comparison with the data reported by Berezutskii [21].

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Appendix. Supplementary data

Supplementary material related to this article can be found online at doi:10.1016/j.calphad.2011.06.001.

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